Concept 5 | Recurrence and Induction

English Student Edition • Focused formulas and guided practice

Formula opener

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1),\ P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = mq'$$

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Worked Solutions **Recurrence and Induction**

Q001 Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 3a_n + (2)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = 8$.
2. $a_3 = 26$.
3. $a_4 = 80$.

Final answer

$a_2 = 8,\ a_3 = 26,\ a_4 = 80$

Q002 Written

For the sequence $\{a_n\}$ with $a_1 = -6$ and recurrence $a_{n+1} = 2a_n + (3)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = -9$.
2. $a_3 = -15$.
3. $a_4 = -27$.

Final answer

$a_2 = -9,\ a_3 = -15,\ a_4 = -27$

Worked Solutions **Recurrence and Induction**

Q003 Written

For the sequence $\{a_n\}$ with $a_1 = -6$ and recurrence $a_{n+1} = 1a_n + (n(n-2))$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = -7$.
2. $a_3 = -7$.
3. $a_4 = -4$.

Final answer

$a_2 = -7,\ a_3 = -7,\ a_4 = -4$

Q004 Written

For the sequence $\{a_n\}$ with $a_1 = -2$ and recurrence $a_{n+1} = 1a_n + (-n)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = -3$.
2. $a_3 = -5$.
3. $a_4 = -8$.

Final answer

$a_2 = -3,\ a_3 = -5,\ a_4 = -8$

Worked Solutions **Recurrence and Induction**

Q005 **Written**

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 1a_n + (2n)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = 4$.
2. $a_3 = 8$.
3. $a_4 = 14$.

Final answer

$a_2 = 4,\ a_3 = 8,\ a_4 = 14$

Q006 **Written**

For the sequence $\{a_n\}$ with $a_1 = -2$ and recurrence $a_{n+1} = 1a_n + (3^n)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = 1$.
2. $a_3 = 10$.
3. $a_4 = 37$.

Final answer

$a_2 = 1,\ a_3 = 10,\ a_4 = 37$

Worked Solutions **Recurrence and Induction**

Q007 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 3a_n + (0)$. The fourth term is:

Solution

1. $a_2 = 9$.
2. $a_3 = 27$.
3. $a_4 = 81$.

Correct option: B

Final answer

81

Q008 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 4a_n + (0)$. The fourth term is:

Solution

1. $a_2 = 8$.
2. $a_3 = 32$.
3. $a_4 = 128$.

Correct option: D

Final answer

128

Worked Solutions **Recurrence and Induction**

Q009 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 2a_n + (n+1)$. The fourth term is:

Solution

1. $a_2 = 6$.
2. $a_3 = 15$.
3. $a_4 = 34$.

Correct option: A

Final answer

34

Q010 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 5$ and recurrence $a_{n+1} = 3a_n + (2n)$. The fourth term is:

Solution

1. $a_2 = 17$.
2. $a_3 = 55$.
3. $a_4 = 171$.

Correct option: B

Final answer

171

Worked Solutions **Recurrence and Induction**

Q011 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 1a_n + (4)$. The fourth term is:

Solution

1. $a_2 = 5$.
2. $a_3 = 9$.
3. $a_4 = 13$.

Correct option: D

Final answer

13

Q012 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = -2a_n + (5)$. The fourth term is:

Solution

1. $a_2 = -3$.
2. $a_3 = 11$.
3. $a_4 = -17$.

Correct option: A

Final answer

-17

Worked Solutions **Recurrence and Induction**

Q013 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = -3a_n + (2)$. The fourth term is:

Solution

1. $a_2 = -1$.
2. $a_3 = 5$.
3. $a_4 = -13$.

Correct option: C

Final answer

-13

Q014 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 1a_n + (n^2)$. The fourth term is:

Solution

1. $a_2 = 5$.
2. $a_3 = 9$.
3. $a_4 = 18$.

Correct option: D

Final answer

18

Worked Solutions **Recurrence and Induction**

Q015 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 2a_n + (1)$. The fourth term is:

Solution

1. $a_2 = 5$.
2. $a_3 = 11$.
3. $a_4 = 23$.

Correct option: C

Final answer

23

Q016 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 5$ and recurrence $a_{n+1} = 1a_n + (-3)$. The fourth term is:

Solution

1. $a_2 = 2$.
2. $a_3 = -1$.
3. $a_4 = -4$.

Correct option: D

Final answer

-4

Worked Solutions **Recurrence and Induction**

Q017 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 1a_n + (n)$. The fourth term is:

Solution

1. $a_2 = 4$.
2. $a_3 = 6$.
3. $a_4 = 9$.

Correct option: B

Final answer

9

Q018 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 2a_n + (-n)$. The fourth term is:

Solution

1. $a_2 = 1$.
2. $a_3 = 0$.
3. $a_4 = -3$.

Correct option: C

Final answer

-3

Worked Solutions **Recurrence and Induction**

Q019 Written

For the sequence $\{a_n\}$ with $a_1 = -1$ and recurrence $a_{n+1} = 1a_n + (5)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 5n - 6$.

Final answer

$a_n = 5n - 6$

Q020 Written

For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 1a_n + (3)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 3n - 2$.

Final answer

$a_n = 3n - 2$

Worked Solutions **Recurrence and Induction**

Q021 Written

For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 1a_n + (-2)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = -2\,(n-3)$.

Final answer

$a_n = -2\,(n-3)$

Q022 Written

For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 1a_n + (6)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 6n - 5$.

Final answer

$a_n = 6n - 5$

Worked Solutions **Recurrence and Induction**

Q023 Written

For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 1a_n + (-5)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 8 - 5n$.

Final answer

$a_n = 8 - 5n$

Q024 Written

For the sequence $\{a_n\}$ with $a_1 = -2$ and recurrence $a_{n+1} = 1a_n + (3)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 3n - 5$.

Final answer

$a_n = 3n - 5$

Worked Solutions **Recurrence and Induction**

Q025 Written

For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 1a_n + (-4)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = -4(n-2)$.

Final answer

$a_n = -4(n-2)$

Q026 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 3a_n + (0)$. The general term is:

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 3^n$.

Correct option: D

Final answer

$a_n = 3^n$

Worked Solutions **Recurrence and Induction**

Q027 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 4a_n + (0)$. The general term is:

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 2^{2n-1}$.

Correct option: A

Final answer

$a_n = 2^{2n-1}$

Q028 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 5a_n + (0)$. The general term is:

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 4\,5^{n-1}$.

Correct option: A

Final answer

$a_n = 4\,5^{n-1}$

Worked Solutions **Recurrence and Induction**

Q029 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 6$ and recurrence $a_{n+1} = 2a_n + (0)$. The general term is:

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 3\,2^n$.

Correct option: A

Final answer

$a_n = 3\,2^n$

Q030 Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 3a_n + (0)$. Find the general term.

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 2\,3^{n-1}$.

Final answer

$a_n = 2\,3^{n-1}$

Worked Solutions **Recurrence and Induction**

Q031 Written

For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 5a_n + (0)$. Find the general term.

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 3\,5^{n-1}$.

Final answer

$a_n = 3\,5^{n-1}$

Q032 Written

For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 3a_n + (0)$. Find the general term.

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 4\,3^{n-1}$.

Final answer

$a_n = 4\,3^{n-1}$

Worked Solutions **Recurrence and Induction**

Q033 Written

For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 4a_n + (0)$. Find the general term.

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 2^{2n-2}$.

Final answer

$a_n = 2^{2n-2}$

Q034 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 1a_n + (4)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 4n - 3$.

Correct option: B

Final answer

$a_n = 4n - 3$

Worked Solutions **Recurrence and Induction**

Q035 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 1a_n + (n^2)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = \frac{2n^3-3n^2+n+24}{6}$.

Correct option: B

Final answer

$$a_n = \frac{2n^3 - 3n^2 + n + 24}{6}$$

Q036 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = -4a_n + (0)$. The general term is:

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = (-4)^{n-1}$.

Correct option: D

Final answer

$$a_n = (-4)^{n-1}$$

Worked Solutions **Recurrence and Induction**

Q037 **MCQ**

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 5$ and recurrence $a_{n+1} = 1a_n + (3^n)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = \frac{3^n+7}{2}$.

Correct option: A

Final answer

$$a_n = \frac{3^n + 7}{2}$$

Q038 **Written**

For the sequence $\{a_n\}$ with $a_1 = -3$ and recurrence $a_{n+1} = -2a_n + (0)$. Find the general term.

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = \frac{3(-2)^n}{2}$.

Final answer

$$a_n = \frac{3\left(-2\right)^n}{2}$$

Worked Solutions **Recurrence and Induction**

Q039 **Written**

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = -5a_n + (0)$. Find the general term.

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 2\left(-5\right)^{n-1}$.

Final answer

$a_n = 2\left(-5\right)^{n-1}$

Q040 **Written**

For the sequence $\{a_n\}$ with $a_1 = -2$ and recurrence $a_{n+1} = -4a_n + (0)$. Find the general term.

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = \frac{\left(-4\right)^n}{2}$.

Final answer

$$a_n = \frac{\left(-4\right)^n}{2}$$

Worked Solutions **Recurrence and Induction**

Q041 *MCQ*

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 5$ and recurrence $a_{n+1} = 1a_n + (-3)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 8 - 3n$.

Correct option: B

Final answer

$a_n = 8 - 3n$

Q042 *MCQ*

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 1a_n + (n)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = \frac{n^2-n+6}{2}$.

Correct option: C

Final answer

$$a_n = \frac{n^2 - n + 6}{2}$$

Worked Solutions **Recurrence and Induction**

Q043 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 1a_n + (-5)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 8 - 5n$.

Correct option: B

Final answer

$a_n = 8 - 5n$

Q044 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 1a_n + (6)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 2\,(3n - 2)$.

Correct option: C

Final answer

$a_n = 2\,(3n - 2)$

Worked Solutions **Recurrence and Induction**

Q045 **Written**

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 3a_n + (-n)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = 5$.
2. $a_3 = 13$.
3. $a_4 = 36$.

Final answer

$a_2 = 5,\ a_3 = 13,\ a_4 = 36$

Q046 **Written**

For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = -2a_n + (5)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = 3$.
2. $a_3 = -1$.
3. $a_4 = 7$.

Final answer

$a_2 = 3,\ a_3 = -1,\ a_4 = 7$

Worked Solutions **Recurrence and Induction**

Q047 **Written**

For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 3a_n + (-n)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = 8$.
2. $a_3 = 22$.
3. $a_4 = 63$.

Final answer

$a_2 = 8,\ a_3 = 22,\ a_4 = 63$

Worked Solutions **Recurrence and Induction**

Q048 Challenge

Solve the third-order recurrence $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$ for $n \geq 3$ with $a_0 = 0, a_1 = 1, a_2 = 6$.

Solution

1. $r^3 - 6r^2 + 11r - 6 = (r-1)(r-2)(r-3)$
2. $a_n = A + B2^n + C3^n$
3. $A = \frac{1}{2},\ B = -2,\ C = \frac{3}{2}$
4. $a_n = -\frac{4\,2^n - 3\,3^n - 1}{2}$

Final answer

$$a_n = -\frac{4\,2^n - 3\,3^n - 1}{2}$$

Worked Solutions **Recurrence and Induction**

Q049 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 1$ and $a_{n+1} = a_n + (2n - 3)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 2k - 3$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 1 + \sum_{k=1}^{n-1}(2k - 3) = (n - 2)^2$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$a_n = (n - 2)^2$

Q050 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 1$ and $a_{n+1} = a_n + (3n)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 3k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 1 + \sum_{k=1}^{n-1}(3k) = \frac{3n^2-3n+2}{2}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{3n^2 - 3n + 2}{2}$$

Worked Solutions **Recurrence and Induction**

Q051 **Written**

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (2n)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 2k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1}(2k) = n^2 - n + 2$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$a_n = n^2 - n + 2$

Q052 **Written**

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (3n^2)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 3k^2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1}(3k^2) = \frac{2n^3-3n^2+n+4}{2}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{2n^3 - 3n^2 + n + 4}{2}$$

Worked Solutions **Recurrence and Induction**

Q053 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (3n + 1)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 3k + 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1}(3k + 1) = \frac{3n^2-n+2}{2}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{3n^2 - n + 2}{2}$$

Q054 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (n(2n + 1))$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k(2k + 1)$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1}(k(2k + 1)) = \frac{4n^3-3n^2-n+12}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{4n^3 - 3n^2 - n + 12}{6}$$

Worked Solutions **Recurrence and Induction**

Q055 MCQ

Select the correct option: If $a_1 = 3$ and $a_{n+1} = a_n + (n^2)$, then a_n is:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k^2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 3 + \sum_{k=1}^{n-1}(k^2) = \frac{2n^3-3n^2+n+18}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: C

Final answer

$$a_n = \frac{2n^3 - 3n^2 + n + 18}{6}$$

Q056 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = a_n + (3n^2)$, then a_n is:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 3k^2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1}(3k^2) = \frac{2n^3-3n^2+n+4}{2}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: A

Final answer

$$a_n = \frac{2n^3 - 3n^2 + n + 4}{2}$$

Worked Solutions **Recurrence and Induction**

Q057 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = a_n + (n^2 + 1)$, then a_n is:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k^2 + 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1}(k^2 + 1) = \frac{2n^3 - 3n^2 + 7n + 6}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: C

Final answer

$$a_n = \frac{2n^3 - 3n^2 + 7n + 6}{6}$$

Q058 MCQ

Select the correct option: If $a_1 = 5$ and $a_{n+1} = a_n + (n(4n-1))$, then a_n is:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k(4k-1)$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 5 + \sum_{k=1}^{n-1}(k(4k-1)) = \frac{(n+1)(8n^2-23n+30)}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: D

Final answer

$$a_n = \frac{(n+1)\left(8n^2 - 23n + 30\right)}{6}$$

Worked Solutions **Recurrence and Induction**

Q059 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = a_n + (8n)$, then a_n is:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 8k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1}(8k) = 2\left(2n^2 - 2n + 1\right)$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: C

Final answer

$a_n = 2\left(2n^2 - 2n + 1\right)$

Q060 MCQ

Select the correct option: find a_4 when $a_1 = 3$ and $a_{n+1} = a_n + (n^2)$:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k^2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 3 + \sum_{k=1}^{n-1}(k^2) = \frac{2n^3 - 3n^2 + n + 18}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: A

Final answer

17

Worked Solutions **Recurrence and Induction**

Q061 MCQ

Select the correct option: find a_5 when $a_1 = 2$ and $a_{n+1} = a_n + (3n^2)$:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 3k^2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1}(3k^2) = \frac{2n^3-3n^2+n+4}{2}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: D

Final answer

92

Q062 MCQ

Select the correct option: find a_3 when $a_1 = 2$ and $a_{n+1} = a_n + (n^2 + 1)$:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k^2 + 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1}(k^2 + 1) = \frac{2n^3-3n^2+7n+6}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: A

Final answer

9

Worked Solutions **Recurrence and Induction**

Q063 MCQ

Select the correct option: find a_6 when $a_1 = 5$ and $a_{n+1} = a_n + (n(4n-1))$:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k(4k-1)$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 5 + \sum_{k=1}^{n-1}(k(4k-1)) = \frac{(n+1)(8n^2-23n+30)}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: B

Final answer

210

Q064 MCQ

Select the correct option: find a_2 when $a_1 = 2$ and $a_{n+1} = a_n + (8n)$:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 8k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1}(8k) = 2\left(2n^2 - 2n + 1\right)$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: A

Final answer

10

Worked Solutions **Recurrence and Induction**

Q065 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (n^2)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k^2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1}(k^2) = \frac{2n^3-3n^2+n+12}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{2n^3 - 3n^2 + n + 12}{6}$$

Q066 MCQ

Select the correct option: If $a_1 = 1$ and $a_{n+1} = a_n + (2n + 1)$, then a_n is:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 2k + 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 1 + \sum_{k=1}^{n-1}(2k + 1) = n^2$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: A

Final answer

$$a_n = n^2$$

Worked Solutions **Recurrence and Induction**

Q067 **Written**

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 1$ and $a_{n+1} = a_n + (2^n)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 2^k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 1 + \sum_{k=1}^{n-1}(2^k) = 2^n - 1$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$a_n = 2^n - 1$

Q068 **Written**

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (-3^n)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = -3^k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1}(-3^k) = -\frac{3^n-7}{2}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = -\frac{3^n - 7}{2}$$

Worked Solutions **Recurrence and Induction**

Q069 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 4$ and $a_{n+1} = a_n + (-4^n)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = -4^k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 4 + \sum_{k=1}^{n-1}(-4^k) = -\frac{4^n-16}{3}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = -\frac{4^n - 16}{3}$$

Worked Solutions **Recurrence and Induction**

Q070 Challenge

A technology startup has a reserve of 400,000 LE, represented by $a_1 = 4$, and burns $0.04n$ hundred-thousand LE each quarter: $a_{n+1} = a_n - 0.04n$. Find an explicit formula for a_n.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = -\frac{k}{25}$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 4 + \sum_{k=1}^{n-1}(-\frac{k}{25}) = -\frac{n^2-n-200}{50}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = -\frac{n^2 - n - 200}{50}$$

Worked Solutions **Recurrence and Induction**

Q071 **Written**

Find the general term of $\{a_n\}$ given $a_1 = 1$ and $a_{n+1} = 3a_n + (4)$.

Solution

1. $c = -2$ from $c = 3c + (4)$.
2. $a_{n+1} - c = 3(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = 3^n - 2$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$a_n = 3^n - 2$

Worked Solutions **Recurrence and Induction**

Q072 **Written**

Find the general term of $\{a_n\}$ given $a_1 = 1$ and $a_{n+1} = 5a_n + (8)$.

Solution

1. $c = -2$ from $c = 5c + (8)$.
2. $a_{n+1} - c = 5(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{3\,5^n - 10}{5}$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{3\,5^n - 10}{5}$$

Worked Solutions **Recurrence and Induction**

Q073 **Written**

Find the general term of $\{a_n\}$ given $a_1 = 2$ and $a_{n+1} = 2a_n + (-1)$.

Solution

1. $c = 1$ from $c = 2c + (-1)$.
2. $a_{n+1} - c = 2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{2^n+2}{2}$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{2^n + 2}{2}$$

Worked Solutions **Recurrence and Induction**

Q074 Written

Find the general term of $\{a_n\}$ given $a_1 = -3$ and $a_{n+1} = 4a_n + (3)$.

Solution

1. $c = -1$ from $c = 4c + (3)$.
2. $a_{n+1} - c = 4(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{4^n+2}{2}$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = -\frac{4^n + 2}{2}$$

Worked Solutions **Recurrence and Induction**

Q075 **Written**

Find the general term of $\{a_n\}$ given $a_1 = 3$ and $a_{n+1} = 3a_n + (-2)$.

Solution

1. $c = 1$ from $c = 3c + (-2)$.
2. $a_{n+1} - c = 3(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{2\,3^n + 3}{3}$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{2\,3^n + 3}{3}$$

Worked Solutions **Recurrence and Induction**

Q076 **Written**

Find the general term of $\{a_n\}$ given $a_1 = -2$ and $a_{n+1} = -3a_n + (4)$.

Solution

1. $c = 1$ from $c = -3c + (4)$.
2. $a_{n+1} - c = -3(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = (-3)^n + 1$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$a_n = (-3)^n + 1$

Worked Solutions **Recurrence and Induction**

Q077 **Written**

Find the general term of $\{a_n\}$ given $a_1 = 1$ and $a_{n+1} = \frac{1}{3}a_n + (2)$.

Solution

1. $c = 3$ from $c = \frac{1}{3}c + (2)$.
2. $a_{n+1} - c = \frac{1}{3}(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = 3\,3^{-n}\,(3^n - 2)$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$a_n = 3\,3^{-n}\,(3^n - 2)$

Worked Solutions **Recurrence and Induction**

Q078 **Written**

Find the general term of $\{a_n\}$ given $a_1 = 2$ and $a_{n+1} = \frac{2}{3}a_n + (-\frac{1}{2})$.

Solution

1. $c = -\frac{3}{2}$ from $c = \frac{2}{3}c + (-\frac{1}{2})$.
2. $a_{n+1} - c = \frac{2}{3}(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{3\,3^{-n}(7\,2^n - 2\,3^n)}{4}$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{3\,3^{-n}\,(7\,2^n - 2\,3^n)}{4}$$

Worked Solutions **Recurrence and Induction**

Q079 Written

Find the general term of $\{a_n\}$ given $a_1 = 4$ and $a_{n+1} = -2a_n + (3)$.

Solution

1. $c = 1$ from $c = -2c + (3)$.
2. $a_{n+1} - c = -2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{3(-2)^n-2}{2}$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = -\frac{3\,(-2)^n - 2}{2}$$

Worked Solutions **Recurrence and Induction**

Q080 MCQ

Select the correct option: If $a_1 = 3$ and $a_{n+1} = -2a_n + (5)$, the general term is:

Solution

1. $c = \frac{5}{3}$ from $c = -2c + (5)$.
2. $a_{n+1} - c = -2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{2(-2)^n - 5}{3}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: B

Final answer

$$a_n = -\frac{2\,(-2)^n - 5}{3}$$

Worked Solutions **Recurrence and Induction**

Q081 MCQ

Select the correct option: If $a_1 = 1$ and $a_{n+1} = \frac{2}{3}a_n + (-2)$, the general term is:

Solution

1. $c = -6$ from $c = \frac{2}{3}c + (-2)$.
2. $a_{n+1} - c = \frac{2}{3}(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{3\,3^{-n}(7\,2^n - 4\,3^n)}{2}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: C

Final answer

$$a_n = \frac{3\,3^{-n}\,(7\,2^n - 4\,3^n)}{2}$$

Worked Solutions **Recurrence and Induction**

Q082 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = 7a_n + (-4)$, the general term is:

Solution

1. $c = \frac{2}{3}$ from $c = 7c + (-4)$.
2. $a_{n+1} - c = 7(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{2(2\,7^n+7)}{21}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: D

Final answer

$$a_n = \frac{2\,(2\,7^n + 7)}{21}$$

Worked Solutions **Recurrence and Induction**

Q083 MCQ

Select the correct option: If $a_1 = 6$ and $a_{n+1} = 2a_n + (-5)$, the general term is:

Solution

1. $c = 5$ from $c = 2c + (-5)$.
2. $a_{n+1} - c = 2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{2^n+10}{2}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: B

Final answer

$$a_n = \frac{2^n + 10}{2}$$

Worked Solutions **Recurrence and Induction**

Q084 MCQ

Select the correct option: If $a_1 = 5$ and $a_{n+1} = -2a_n + (7)$, the general term is:

Solution

1. $c = \frac{7}{3}$ from $c = -2c + (7)$.
2. $a_{n+1} - c = -2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{4(-2)^n - 7}{3}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: C

Final answer

$$a_n = -\frac{4\,(-2)^n - 7}{3}$$

Worked Solutions **Recurrence and Induction**

Q085 *MCQ*

Select the correct option: If $a_1 = 4$ and $a_{n+1} = -5a_n + (-2)$, the general term is:

Solution

1. $c = -\frac{1}{3}$ from $c = -5c + (-2)$.
2. $a_{n+1} - c = -5(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{13(-5)^n+5}{15}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: D

Final answer

$$a_n = -\frac{13\,(-5)^n + 5}{15}$$

Worked Solutions **Recurrence and Induction**

Q086 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = -3a_n + (5)$, the general term is:

Solution

1. $c = \frac{5}{4}$ from $c = -3c + (5)$.
2. $a_{n+1} - c = -3(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{(-3)^n - 5}{4}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: B

Final answer

$$a_n = -\frac{(-3)^n - 5}{4}$$

Worked Solutions Recurrence and Induction

Q087 MCQ

Select the correct option: If $a_1 = 1$ and $a_{n+1} = \frac{1}{2}a_n + (7)$, the general term is:

Solution

1. $c = 14$ from $c = \frac{1}{2}c + (7)$.
2. $a_{n+1} - c = \frac{1}{2}(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = 2\,2^{-n}\,(7\,2^n - 13)$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: C

Final answer

$a_n = 2\,2^{-n}\,(7\,2^n - 13)$

Worked Solutions **Recurrence and Induction**

Q088 MCQ

Select the correct option: If $a_1 = 1$ and $a_{n+1} = 2a_n + (-1)$, the general term is:

Solution

1. $c = 1$ from $c = 2c + (-1)$.
2. $a_{n+1} - c = 2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = 1$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: D

Final answer

$a_n = 1$

Worked Solutions **Recurrence and Induction**

Q089 MCQ

Select the correct option: If $a_1 = 3$ and $a_{n+1} = -2a_n + (5)$, the general term is:

Solution

1. $c = \frac{5}{3}$ from $c = -2c + (5)$.
2. $a_{n+1} - c = -2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{2(-2)^n - 5}{3}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: D

Final answer

$$a_n = -\frac{2\left(-2\right)^n - 5}{3}$$

Worked Solutions **Recurrence and Induction**

Q090 Challenge

A sequence satisfies $a_{n+1} = 4a_n - 9$ with $a_1 = 2$. Find the equilibrium value, the explicit formula, and a_5.

Solution

1. $c = 3$ from $c = 4c + (-9)$.
2. $a_{n+1} - c = 4(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{2^{2n}-12}{4}$.
5. Substitute $n = 1$ to verify the given first term.
6. $a_5 = -253$

Final answer

$$c = 3, \quad a_n = -\frac{2^{2n} - 12}{4}, \quad a_5 = -253$$

Worked Solutions **Recurrence and Induction**

Q091 **Written**

Use mathematical induction to prove $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$.

Solution

1. [1] Base case $n = 1$: both sides equal 1.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^{k} i = \frac{k(k+1)}{2}$.
3. For $n = k + 1$, add the next term n to the left side and use the hypothesis.
4. $\frac{k(k+1)}{2} + n = \frac{(k+1)(k+2)}{2}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$\frac{n\,(n+1)}{2}$$

Worked Solutions **Recurrence and Induction**

Q092 **Written**

Use mathematical induction to prove $\sum_{i=1}^{n}(2i-1)^2 = \frac{n(2n-1)(2n+1)}{3}$.

Solution

1. [1] Base case $n = 1$: both sides equal 1.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^{k}(2i-1)^2 = \frac{k(2k-1)(2k+1)}{3}$.
3. For $n = k + 1$, add the next term $(2k+1)^2$ to the left side and use the hypothesis.
4. $\frac{k(2k-1)(2k+1)}{3} + (2k+1)^2 = \frac{(k+1)(2k+1)(2k+3)}{3}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$\frac{n\,(2n-1)\,(2n+1)}{3}$$

Worked Solutions **Recurrence and Induction**

Q093 **Written**

Use mathematical induction to prove $\sum_{i=1}^{n} i\,(i+1) = \frac{n(n+1)(n+2)}{3}$.

Solution

1. [1] Base case $n = 1$: both sides equal 2.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^{k} i\,(i+1) = \frac{k(k+1)(k+2)}{3}$.
3. For $n = k + 1$, add the next term $(k+1)\,(k+2)$ to the left side and use the hypothesis.
4. $\frac{k(k+1)(k+2)}{3} + (k+1)\,(k+2) = \frac{(k+1)(k+2)(k+3)}{3}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$\frac{n\,(n+1)\,(n+2)}{3}$$

Worked Solutions **Recurrence and Induction**

Q094 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^{n} i\,(i+1)$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 2.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^{k} i\,(i+1) = \frac{k(k+1)(k+2)}{3}$.
3. For $n = k+1$, add the next term $(k+1)\,(k+2)$ to the left side and use the hypothesis.
4. $\frac{k(k+1)(k+2)}{3} + (k+1)\,(k+2) = \frac{(k+1)(k+2)(k+3)}{3}$, which is the right side at $k+1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: A

Final answer

$$\frac{n\,(n+1)\,(n+2)}{3}$$

Worked Solutions **Recurrence and Induction**

Q095 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^{n} 3 \cdot 4^{i-1}$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 3.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^{k} 3 \cdot 4^{i-1} = 4^k - 1$.
3. For $n = k + 1$, add the next term $3\,4^k$ to the left side and use the hypothesis.
4. $4^k - 1 + 3\,4^k = \left(2\,2^k - 1\right)\left(2\,2^k + 1\right)$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: B

Final answer

$4^n - 1$

Q096 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^{n} 2i + 3$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 5.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^{k} 2i + 3 = k\,(k + 4)$.
3. For $n = k + 1$, add the next term $2k + 5$ to the left side and use the hypothesis.
4. $k\,(k + 4) + 2k + 5 = (k + 1)\,(k + 5)$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: C

Final answer

$n\,(n + 4)$

Worked Solutions **Recurrence and Induction**

Q097 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^{n} 2i^2 + 3i$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 5.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^{k} 2i^2 + 3i = \frac{k(k+1)(4k+11)}{6}$.
3. For $n = k + 1$, add the next term $(k + 1)(2k + 5)$ to the left side and use the hypothesis.
4. $\frac{k(k+1)(4k+11)}{6} + (k + 1)(2k + 5) = \frac{(k+1)(k+2)(4k+15)}{6}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: B

Final answer

$$\frac{n(n+1)(4n+11)}{6}$$

Worked Solutions **Recurrence and Induction**

Q098 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^{n} 3i^2 + 2i$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 5.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^{k} 3i^2 + 2i = \frac{k(k+1)(2k+3)}{2}$.
3. For $n = k + 1$, add the next term $(k + 1)(3k + 5)$ to the left side and use the hypothesis.
4. $\frac{k(k+1)(2k+3)}{2} + (k + 1)(3k + 5) = \frac{(k+1)(k+2)(2k+5)}{2}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: C

Final answer

$$\frac{n(n+1)(2n+3)}{2}$$

Worked Solutions **Recurrence and Induction**

Q099 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^{n} 7 \cdot 2^{i-1}$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 7.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^{k} 7 \cdot 2^{i-1} = 7\left(2^k - 1\right)$.
3. For $n = k + 1$, add the next term $7\,2^k$ to the left side and use the hypothesis.
4. $7\left(2^k - 1\right) + 7\,2^k = 7\left(2\,2^k - 1\right)$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: A

Final answer

$7\,(2^n - 1)$

Worked Solutions **Recurrence and Induction**

Q100 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^{n} i\,(i+1)$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 2.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^{k} i\,(i+1) = \frac{k(k+1)(k+2)}{3}$.
3. For $n = k+1$, add the next term $(k+1)\,(k+2)$ to the left side and use the hypothesis.
4. $\frac{k(k+1)(k+2)}{3} + (k+1)\,(k+2) = \frac{(k+1)(k+2)(k+3)}{3}$, which is the right side at $k+1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: C

Final answer

$$\frac{n\,(n+1)\,(n+2)}{3}$$

Worked Solutions **Recurrence and Induction**

Q101 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^{n} 3 \cdot 4^{i-1}$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 3.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^{k} 3 \cdot 4^{i-1} = 4^k - 1$.
3. For $n = k + 1$, add the next term $3\,4^k$ to the left side and use the hypothesis.
4. $4^k - 1 + 3\,4^k = \left(2\,2^k - 1\right)\left(2\,2^k + 1\right)$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: A

Final answer

$4^n - 1$

Worked Solutions **Recurrence and Induction**

Q102 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^{n} 2i + 3$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 5.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^{k} 2i + 3 = k(k+4)$.
3. For $n = k + 1$, add the next term $2k + 5$ to the left side and use the hypothesis.
4. $k(k+4) + 2k + 5 = (k+1)(k+5)$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: B

Final answer

$n(n+4)$

Worked Solutions **Recurrence and Induction**

Q103 Written

Use mathematical induction to prove $\sum_{i=1}^{n}(2i-1)=n^2$.

Solution

1. [1] Base case $n=1$: both sides equal 1.
2. [2] Assume the statement holds for $n=k$: $\sum_{i=1}^{k}(2i-1)=k^2$.
3. For $n=k+1$, add the next term $2k+1$ to the left side and use the hypothesis.
4. $k^2+2k+1=(k+1)^2$, which is the right side at $k+1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

n^2

Worked Solutions **Recurrence and Induction**

Q104 Written

Use mathematical induction to prove $\sum_{i=1}^{n} i(i+1) = \frac{n(n+1)(n+2)}{3}$.

Solution

1. [1] Base case $n = 1$: both sides equal 2.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^{k} i(i+1) = \frac{k(k+1)(k+2)}{3}$.
3. For $n = k + 1$, add the next term $(k+1)(k+2)$ to the left side and use the hypothesis.
4. $\frac{k(k+1)(k+2)}{3} + (k+1)(k+2) = \frac{(k+1)(k+2)(k+3)}{3}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$\frac{n(n+1)(n+2)}{3}$$

Worked Solutions **Recurrence and Induction**

Q105 **Written**

Use mathematical induction to prove $\sum_{i=1}^{n} i(i+2) = \frac{n(n+1)(2n+7)}{6}$.

Solution

1. [1] Base case $n = 1$: both sides equal 3.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^{k} i(i+2) = \frac{k(k+1)(2k+7)}{6}$.
3. For $n = k+1$, add the next term $(k+1)(k+3)$ to the left side and use the hypothesis.
4. $\frac{k(k+1)(2k+7)}{6} + (k+1)(k+3) = \frac{(k+1)(k+2)(2k+9)}{6}$, which is the right side at $k+1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$\frac{n(n+1)(2n+7)}{6}$$

Worked Solutions **Recurrence and Induction**

Q106 **Written**

Use mathematical induction to prove $\sum_{i=1}^{n} 3\,4^{i-1} = 4^n - 1$.

Solution

1. [1] Base case $n = 1$: both sides equal 3.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^{k} 3\,4^{i-1} = 4^k - 1$.
3. For $n = k + 1$, add the next term $3\,4^k$ to the left side and use the hypothesis.
4. $4^k - 1 + 3\,4^k = \left(2\,2^k - 1\right)\left(2\,2^k + 1\right)$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$4^n - 1$

Worked Solutions **Recurrence and Induction**

Q107 Written

Use mathematical induction to prove $\sum_{i=0}^{n} 10^i = \frac{10\,10^n - 1}{9}$.

Solution

1. [1] Base case $n = 1$: both sides equal 11.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=0}^{k} 10^i = \frac{10\,10^k - 1}{9}$.
3. For $n = k + 1$, add the next term 10^{k+1} to the left side and use the hypothesis.
4. $\frac{10\,10^k - 1}{9} + 10^{k+1} = \frac{100\,10^k - 1}{9}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$\frac{10\,10^n - 1}{9}$$

Worked Solutions **Recurrence and Induction**

Q108 **Challenge**

Use mathematical induction to prove $\sum_{i=1}^{n}(i^2+i+1)=\frac{n(n^2+3n+5)}{3}$.

Solution

1. [1] Base case $n=1$: both sides equal 3.
2. [2] Assume the statement holds for $n=k$: $\sum_{i=1}^{k}(i^2+i+1)=\frac{k(k^2+3k+5)}{3}$.
3. For $n=k+1$, add the next term k^2+3k+3 to the left side and use the hypothesis.
4. $\frac{k(k^2+3k+5)}{3}+k^2+3k+3=\frac{(k+1)(k^2+5k+9)}{3}$, which is the right side at $k+1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$\frac{n\left(n^2+3n+5\right)}{3}$$

Q109 Written

Prove by mathematical induction that $2^n > 3n$ for $n \geq 4$.

Solution

1. $n = 4:\quad 2^4 = 16 > 12 = 12.$
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $2^k + 1 > 2(3k + 0)$ after multiplying by the positive base.
4. $2(3k + 0) - (3(k + 1) + 0) = 3\,(k - 1)$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Final answer

$2^n > 3n$

Q110 Written

Prove by mathematical induction that $3^n > 5n + 1$ for $n \geq 3$.

Solution

1. $n = 3:\quad 3^3 = 27 > 16 = 16.$
2. Assume the inequality holds for $n = k$ with $k \geq 3$.
3. $3^k + 1 > 3(5k + 1)$ after multiplying by the positive base.
4. $3(5k + 1) - (5(k + 1) + 1) = 10k - 3$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Final answer

$3^n > 5n + 1$

Worked Solutions **Recurrence and Induction**

Q111 Written

Prove by mathematical induction that $3^n > 2n$ for $n \geq 1$.

Solution

1. $n = 1:\quad 3^1 = 3 > 2 = 2.$
2. Assume the inequality holds for $n = k$ with $k \geq 1$.
3. $3^k + 1 > 3(2k + 0)$ after multiplying by the positive base.
4. $3(2k + 0) - (2(k + 1) + 0) = 2\,(2k - 1)$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Final answer

$3^n > 2n$

Worked Solutions **Recurrence and Induction**

Q112 **Written**

Prove by mathematical induction that $2^n > 1n + 2$ for $n \geq 3$.

Solution

1. $n = 3:\quad 2^3 = 8 > 5 = 5.$
2. Assume the inequality holds for $n = k$ with $k \geq 3$.
3. $2^k + 1 > 2(1k + 2)$ after multiplying by the positive base.
4. $2(1k + 2) - (1(k + 1) + 2) = k + 1$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Final answer

$2^n > 1n + 2$

Worked Solutions **Recurrence and Induction**

Q113 **Written**

Prove by mathematical induction that $3^n > 5n + 1$ for $n \geq 4$.

Solution

1. $n = 4:\quad 3^4 = 81 > 21 = 21.$
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $3^k + 1 > 3(5k + 1)$ after multiplying by the positive base.
4. $3(5k + 1) - (5(k + 1) + 1) = 10k - 3$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Final answer

$3^n > 5n + 1$

Worked Solutions **Recurrence and Induction**

Q114 **Written**

Prove by mathematical induction that $3^n > 5n + 2$ for $n \geq 4$.

Solution

1. $n = 4:\quad 3^4 = 81 > 22 = 22.$
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $3^k + 1 > 3(5k + 2)$ after multiplying by the positive base.
4. $3(5k + 2) - (5(k + 1) + 2) = 10k - 1$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Final answer

$3^n > 5n + 2$

Worked Solutions **Recurrence and Induction**

Q115 Written

Prove by mathematical induction that $4^n > 2n + 3$ for $n \geq 4$.

Solution

1. $n = 4:\quad 4^4 = 256 > 11 = 11.$
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $4^k + 1 > 4(2k + 3)$ after multiplying by the positive base.
4. $4(2k + 3) - (2(k + 1) + 3) = 6k + 7$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Final answer

$4^n > 2n + 3$

Worked Solutions **Recurrence and Induction**

Q116 MCQ

Select the correct option: which inequality is proved for $n \geq 3$?:

Solution

1. $n = 3: \quad 2^3 = 8 > 5 = 5.$
2. Assume the inequality holds for $n = k$ with $k \geq 3$.
3. $2^k + 1 > 2(1k + 2)$ after multiplying by the positive base.
4. $2(1k + 2) - (1(k + 1) + 2) = k + 1$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Correct option: D

Final answer

$2^n > 1n + 2$

Worked Solutions **Recurrence and Induction**

Q117 MCQ

Select the correct option: which inequality is proved for $n \geq 2$?:

Solution

1. $n = 2: \quad 3^2 = 9 > 4 = 4.$
2. Assume the inequality holds for $n = k$ with $k \geq 2$.
3. $3^k + 1 > 3(2k + 0)$ after multiplying by the positive base.
4. $3(2k + 0) - (2(k + 1) + 0) = 2\,(2k - 1)$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Correct option: A

Final answer

$3^n > 2n$

Worked Solutions **Recurrence and Induction**

Q118 MCQ

Select the correct option: which inequality is proved for $n \geq 4$?:

Solution

1. $n = 4: \quad 3^4 = 81 > 21 = 21.$
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $3^k + 1 > 3(5k + 1)$ after multiplying by the positive base.
4. $3(5k + 1) - (5(k + 1) + 1) = 10k - 3$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Correct option: B

Final answer

$3^n > 5n + 1$

Worked Solutions **Recurrence and Induction**

Q119 MCQ

Select the correct option: which inequality is proved for $n \geq 3$?:

Solution

1. $n = 3 : \quad 3^3 = 27 > 9 = 9.$
2. Assume the inequality holds for $n = k$ with $k \geq 3$.
3. $3^k + 1 > 3(3k + 0)$ after multiplying by the positive base.
4. $3(3k + 0) - (3(k + 1) + 0) = 3\,(2k - 1)$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Correct option: D

Final answer

$3^n > 3n$

Worked Solutions **Recurrence and Induction**

Q120 MCQ

Select the correct option: which inequality is proved for $n \geq 4$?:

Solution

1. $n = 4: \quad 3^4 = 81 > 22 = 22.$
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $3^k + 1 > 3(5k + 2)$ after multiplying by the positive base.
4. $3(5k + 2) - (5(k + 1) + 2) = 10k - 1$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Correct option: A

Final answer

$3^n > 5n + 2$

Worked Solutions **Recurrence and Induction**

Q121 MCQ

Select the correct option: which inequality is proved for $n \geq 3$?:

Solution

1. $n = 3: \quad 4^3 = 64 > 10 = 10.$
2. Assume the inequality holds for $n = k$ with $k \geq 3$.
3. $4^k + 1 > 4(3k + 1)$ after multiplying by the positive base.
4. $4(3k + 1) - (3(k + 1) + 1) = 9k$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Correct option: B

Final answer

$4^n > 3n + 1$

Worked Solutions **Recurrence and Induction**

Q122 MCQ

Select the correct option: which inequality is proved for $n \geq 4$?:

Solution

1. $n = 4: \quad 4^4 = 256 > 11 = 11.$
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $4^k + 1 > 4(2k + 3)$ after multiplying by the positive base.
4. $4(2k + 3) - (2(k + 1) + 3) = 6k + 7$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Correct option: D

Final answer

$4^n > 2n + 3$

Worked Solutions **Recurrence and Induction**

Q123 MCQ

Select the correct option: which inequality is proved for $n \geq 4$?:

Solution

1. $n = 4: \quad 4^4 = 256 > 17 = 17.$
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $4^k + 1 > 4(4k + 1)$ after multiplying by the positive base.
4. $4(4k + 1) - (4(k + 1) + 1) = 12k - 1$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Correct option: A

Final answer

$4^n > 4n + 1$

Worked Solutions **Recurrence and Induction**

Q124 **Written**

Prove by mathematical induction that $2^n > 3n + 1$ for $n \geq 4$.

Solution

1. $n = 4:$ $2^4 = 16 > 13 = 13.$
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $2^k + 1 > 2(3k + 1)$ after multiplying by the positive base.
4. $2(3k + 1) - (3(k + 1) + 1) = 3k - 2$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Final answer

$2^n > 3n + 1$

Worked Solutions **Recurrence and Induction**

Q125 **MCQ**

Select the correct option: which inequality is proved for $n \geq 3$?:

Solution

1. $n = 3: \quad 4^3 = 64 > 9 = 9.$
2. Assume the inequality holds for $n = k$ with $k \geq 3$.
3. $4^k + 1 > 4(3k + 0)$ after multiplying by the positive base.
4. $4(3k + 0) - (3(k + 1) + 0) = 3\,(3k - 1)$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Correct option: B

Final answer

$4^n > 3n$

Worked Solutions **Recurrence and Induction**

Q126 **Challenge**

When $h > 0$ and $n \geq 3$, prove $(1+h)^n > 1 + nh^2$.

Solution

1. $n = 3:\quad (1+h)^3 = 1 + 3h + 3h^2 + h^3 > 1 + 3h^2$ because h>0 and $3h + h^3 > 0$.
2. Assume $(1+h)^k > 1 + kh^2$ for $k \geq 3$.
3. Multiplying by the positive factor $1 + h$ gives a lower bound $(1+h)(1+kh^2)$.
4. $(1+h)(1+kh^2) - [1 + (k+1)h^2] = h(1 - h + kh^2)$. The quadratic $1 - h + kh^2$ is positive for $k \geq 3$ because its discriminant is $1 - 4k < 0$.
5. Therefore the inequality holds for every $n \geq 3$ and $h > 0$.

Final answer

$(1+h)^n > 1 + nh^2$

Worked Solutions **Recurrence and Induction**

Q127 Written

Prove that $n^3 + (n+1)^3 + (n+2)^3$ is a multiple of 9 for every natural number n, using mathematical induction.

Solution

1. [1] Base case: $n = 1, \quad f(1) = 36 = 9(4)$, so the statement holds.
2. [2] Assume $f(k) = 9m$ for an integer m.
3. [3] The induction difference is $f(k+1) - f(k) = 9\left(k^2 + 3k + 3\right) = 9(k^2 + 3k + 3)$.
4. [4] Therefore $f(k+1) = 9\left[m + k^2 + 3k + 3\right]$ is a multiple of the divisor.
5. The result follows for every natural number by mathematical induction.

Final answer

$9 \mid [n^3 + (n+1)^3 + (n+2)^3]$

Worked Solutions **Recurrence and Induction**

Q128 **Written**

Prove that $n^3 + 2n$ is a multiple of 3 for every natural number n, using mathematical induction.

Solution

1. [1] Base case: $n = 1, \quad f(1) = 3 = 3(1)$, so the statement holds.
2. [2] Assume $f(k) = 3m$ for an integer m.
3. [3] The induction difference is $f(k+1) - f(k) = 3\left(k^2 + k + 1\right) = 3(k^2 + k + 1)$.
4. [4] Therefore $f(k+1) = 3\left[m + k^2 + k + 1\right]$ is a multiple of the divisor.
5. The result follows for every natural number by mathematical induction.

Final answer

$3 \mid (n^3 + 2n)$

Worked Solutions **Recurrence and Induction**

Q129 **Written**

Prove by mathematical induction that $n^2 + n$ is a multiple of 2.

Solution

1. [1] Base case: $n = 1, \quad f(1) = 2 = 2(1)$.
2. [2] Assume $f(k) = 2m$ for an integer m.
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2\prod_{j=1}^{1}(k+0+j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Final answer

$2 \mid (n^2 + n)$

Worked Solutions **Recurrence and Induction**

Q130 **Written**

Prove by mathematical induction that $n^2 + 5n + 8$ is a multiple of 2.

Solution

1. [1] Base case: $n = 1, \quad f(1) = 14 = 2(7)$.
2. [2] Assume $f(k) = 2m$ for an integer m.
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2\prod_{j=1}^{1}(k+2+j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Final answer

$2 \mid (n^2 + 5n + 8)$

Worked Solutions **Recurrence and Induction**

Q131 **Written**

Prove by mathematical induction that $n^2 + 7n + 12$ is a multiple of 2.

Solution

1. [1] Base case: $n = 1, \quad f(1) = 20 = 2(10)$.
2. [2] Assume $f(k) = 2m$ for an integer m.
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2\prod_{j=1}^{1}(k+3+j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Final answer

$2 \mid (n^2 + 7n + 12)$

Worked Solutions **Recurrence and Induction**

Q132 **Written**

Prove by mathematical induction that $n^2 + 11n + 32$ is a multiple of 2.

Solution

1. [1] Base case: $n = 1, \quad f(1) = 44 = 2(22)$.
2. [2] Assume $f(k) = 2m$ for an integer m.
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2 \prod_{j=1}^{1}(k+5+j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Final answer

$2 \mid (n^2 + 11n + 32)$

Worked Solutions **Recurrence and Induction**

Q133 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + n$ for every natural number n?:

Solution

1. [1] Base case: $n = 1, \quad f(1) = 2 = 2(1)$.
2. [2] Assume $f(k) = 2m$ for an integer m.
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2\prod_{j=1}^{1}(k + 0 + j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Correct option: A

Final answer

2

Worked Solutions **Recurrence and Induction**

Q134 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + 5n + 8$ for every natural number n?:

Solution

1. [1] Base case: $n = 1, \quad f(1) = 14 = 2(7)$.
2. [2] Assume $f(k) = 2m$ for an integer m.
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2\prod_{j=1}^{1}(k+2+j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Correct option: D

Final answer

2

Worked Solutions **Recurrence and Induction**

Q135 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + 7n + 12$ for every natural number n?:

Solution

1. [1] Base case: $n = 1, \quad f(1) = 20 = 2(10)$.
2. [2] Assume $f(k) = 2m$ for an integer m.
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2\prod_{j=1}^{1}(k+3+j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Correct option: C

Final answer

2

Worked Solutions **Recurrence and Induction**

Q136 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + 11n + 32$ for every natural number n?:

Solution

1. [1] Base case: $n = 1, \quad f(1) = 44 = 2(22)$.
2. [2] Assume $f(k) = 2m$ for an integer m.
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2 \prod_{j=1}^{1}(k+5+j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Correct option: C

Final answer

2

Worked Solutions **Recurrence and Induction**

Q137 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^3 + 3n^2 + 2n$ for every natural number n?:

Solution

1. [1] Base case: $n = 1, \quad f(1) = 6 = 6(1)$.
2. [2] Assume $f(k) = 6m$ for an integer m.
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 3\prod_{j=1}^{2}(k + 0 + j)$.
4. [4] Since the remaining 2 consecutive integers contain a multiple of 2 factorial, the difference is a multiple of 6.
5. Thus $f(k+1) = 6[m + \text{an integer}]$ and the result follows by induction.

Correct option: B

Final answer

6

Worked Solutions · **Recurrence and Induction**

Q138 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^3 + 9n^2 + 26n + 30$ for every natural number n?:

Solution

1. [1] Base case: $n = 1, \quad f(1) = 66 = 6(11)$.
2. [2] Assume $f(k) = 6m$ for an integer m.
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 3\prod_{j=1}^{2}(k+2+j)$.
4. [4] Since the remaining 2 consecutive integers contain a multiple of 2 factorial, the difference is a multiple of 6.
5. Thus $f(k+1) = 6[m + \text{an integer}]$ and the result follows by induction.

Correct option: A

Final answer

6

Worked Solutions **Recurrence and Induction**

Q139 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^3 + 12n^2 + 47n + 60$ for every natural number n?:

Solution

1. [1] Base case: $n = 1, \quad f(1) = 120 = 6(20)$.
2. [2] Assume $f(k) = 6m$ for an integer m.
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 3\prod_{j=1}^{2}(k+3+j)$.
4. [4] Since the remaining 2 consecutive integers contain a multiple of 2 factorial, the difference is a multiple of 6.
5. Thus $f(k+1) = 6[m + \text{an integer}]$ and the result follows by induction.

Correct option: D

Final answer

6

Worked Solutions **Recurrence and Induction**

Q140 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^3 + 18n^2 + 107n + 216$ for every natural number n?:

Solution

1. [1] Base case: $n = 1, \quad f(1) = 342 = 6(57)$.
2. [2] Assume $f(k) = 6m$ for an integer m.
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 3\prod_{j=1}^{2}(k+5+j)$.
4. [4] Since the remaining 2 consecutive integers contain a multiple of 2 factorial, the difference is a multiple of 6.
5. Thus $f(k+1) = 6[m + \text{an integer}]$ and the result follows by induction.

Correct option: D

Final answer

6

Worked Solutions **Recurrence and Induction**

Q141 **MCQ**

Select the correct option: which positive integer is guaranteed to divide $n^2 + n + 2$ for every natural number n?:

Solution

1. [1] Base case: $n = 1, \quad f(1) = 4 = 2(2)$.
2. [2] Assume $f(k) = 2m$ for an integer m.
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2\prod_{j=1}^{1}(k+0+j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Correct option: D

Final answer

2

Worked Solutions **Recurrence and Induction**

Q142 **Written**

Prove by mathematical induction that $2n^3 - 3n^2 + n$ is a multiple of 6.

Solution

1. [1] Base case: $n = 1, \quad f(1) = 0 = 6(0)$, so the statement holds.
2. [2] Assume $f(k) = 6m$ for an integer m.
3. [3] The induction difference is $f(k+1) - f(k) = 6k^2 = 6(k^2)$.
4. [4] Therefore $f(k+1) = 6\left[m + k^2\right]$ is a multiple of the divisor.
5. The result follows for every natural number by mathematical induction.

Final answer

$6 \mid (2n^3 - 3n^2 + n)$

Worked Solutions **Recurrence and Induction**

Q143 **Written**

Prove by mathematical induction that $4n^3 - n$ is a multiple of 3.

Solution

1. [1] Base case: $n = 1, \quad f(1) = 3 = 3(1)$, so the statement holds.
2. [2] Assume $f(k) = 3m$ for an integer m.
3. [3] The induction difference is $f(k+1) - f(k) = 3\,(2k+1)^2 = 3((2k+1)^2)$.
4. [4] Therefore $f(k+1) = 3\left[m + (2k+1)^2\right]$ is a multiple of the divisor.
5. The result follows for every natural number by mathematical induction.

Final answer

$3 \mid (4n^3 - n)$

Worked Solutions **Recurrence and Induction**

Q144 **Challenge**

For any integer $n \geq 1$, prove that $(n+1)^n - 1$ is a multiple of n^2.

Solution

1. Use the binomial theorem on $(n+1)^n = (1+n)^n$.
2. $(1+n)^n - 1 = \binom{n}{1}n + \binom{n}{2}n^2 + \cdots + n^n$.
3. $\binom{n}{1}n = n^2$, and every remaining term also contains n^2.
4. $(n+1)^n - 1 = n^2[1 + \binom{n}{2} + \binom{n}{3}n + \cdots + n^{n-2}]$. The bracket is an integer for every integer $n \geq 1$.
5. Hence the expression is a multiple of n^2.

Final answer

$n^2 \mid [(n+1)^n - 1]$

Worked Solutions **Recurrence and Induction**

Q145 Written

For the sequence determined by $a_1 = 3$ and $a_{n+1} = \frac{a_n^2-1}{n+1}$, find a_2, a_3, a_4 and estimate the general term.

Solution

1. $a_2 = 4, \quad a_3 = 5, \quad a_4 = 6.$
2. The pattern suggests $a_n = n + 2.$

Final answer

$a_2 = 4, \quad a_3 = 5, \quad a_4 = 6; \quad a_n = n + 2$

Q146 Written

For the sequence determined by $a_1 = -1$ and $a_{n+1} = a_n^2 + 2na_n - 2$, find a_2, a_3, a_4 and estimate the general term.

Solution

1. $a_2 = -3, \quad a_3 = -5, \quad a_4 = -7.$
2. The pattern suggests $a_n = 1 - 2n.$

Final answer

$a_2 = -3, \quad a_3 = -5, \quad a_4 = -7; \quad a_n = 1 - 2n$

Worked Solutions · **Recurrence and Induction**

Q147 Written

For the sequence determined by $a_1 = 1$ and $a_{n+1} = a_n^2 - n^2 a_n + (n+1)^2$, find a_2, a_3, a_4 and estimate the general term.

Solution

1. $a_2 = 4, \quad a_3 = 9, \quad a_4 = 16.$
2. The pattern suggests $a_n = n^2$.

Final answer

$a_2 = 4, \quad a_3 = 9, \quad a_4 = 16; \quad a_n = n^2$

Q148 Written

For the sequence determined by $a_1 = 1$ and $a_{n+1} = \frac{a_n - 4}{a_n - 3}$, find a_2, a_3, a_4 and estimate the general term.

Solution

1. $a_2 = \frac{3}{2}, \quad a_3 = \frac{5}{3}, \quad a_4 = \frac{7}{4}.$
2. The pattern suggests $a_n = \frac{2n-1}{n}$.

Final answer

$$a_2 = \frac{3}{2}, \quad a_3 = \frac{5}{3}, \quad a_4 = \frac{7}{4}; \quad a_n = \frac{2n-1}{n}$$

Worked Solutions **Recurrence and Induction**

Q149 **Written**

For a sequence with $a_1 = 3$ and $a_{n+1} = \frac{a_n^2 - 1}{n+1}$, prove by mathematical induction that $a_n = n + 2$.

Solution

1. [1] Base case: $a_1 = 3 = 3$.
2. [2] Assume $a_k = k + 2$ for some positive integer k.
3. [3] Substitute into $a_{k+1} = \frac{{a_k}^2 - 1}{k+1}$: $a_{k+1} = k + 3$.
4. [4] This is $k + 3$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Final answer

$a_n = n + 2$

Worked Solutions **Recurrence and Induction**

Q150 MCQ

Select the correct option: for $a_1 = 1$ and $a_{n+1} = a_n + 2$, the general term is:

Solution

1. [1] Base case: $a_1 = 1 = 1$.
2. [2] Assume $a_k = 2k - 1$ for some positive integer k.
3. [3] Substitute into $a_{k+1} = a_k + 2$: $a_{k+1} = 2k + 1$.
4. [4] This is $2k + 1$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: D

Final answer

$a_n = 2n - 1$

Worked Solutions **Recurrence and Induction**

Q151 MCQ

Select the correct option: for $a_1 = 2$ and $a_{n+1} = a_n + 2$, the general term is:

Solution

1. [1] Base case: $a_1 = 2 = 2$.
2. [2] Assume $a_k = 2k$ for some positive integer k.
3. [3] Substitute into $a_{k+1} = a_k + 2$: $a_{k+1} = 2(k + 1)$.
4. [4] This is $2(k + 1)$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: C

Final answer

$a_n = 2n$

Worked Solutions **Recurrence and Induction**

Q152 MCQ

Select the correct option: for $a_1 = 3$ and $a_{n+1} = a_n + 2$, the general term is:

Solution

1. [1] Base case: $a_1 = 3 = 3$.
2. [2] Assume $a_k = 2k + 1$ for some positive integer k.
3. [3] Substitute into $a_{k+1} = a_k + 2$: $a_{k+1} = 2k + 3$.
4. [4] This is $2k + 3$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: B

Final answer

$a_n = 2n + 1$

Worked Solutions **Recurrence and Induction**

Q153 **MCQ**

Select the correct option: for $a_1 = 4$ and $a_{n+1} = a_n + 2$, the general term is:

Solution

1. [1] Base case: $a_1 = 4 = 4$.
2. [2] Assume $a_k = 2(k+1)$ for some positive integer k.
3. [3] Substitute into $a_{k+1} = a_k + 2$: $a_{k+1} = 2(k+2)$.
4. [4] This is $2(k+2)$, the proposed formula with $n = k+1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: A

Final answer

$a_n = 2(n+1)$

Q154 MCQ

Select the correct option: for $a_1 = 1$ and $a_{n+1} = a_n + 3$, the general term is:

Solution

1. [1] Base case: $a_1 = 1 = 1$.
2. [2] Assume $a_k = 3k - 2$ for some positive integer k.
3. [3] Substitute into $a_{k+1} = a_k + 3$: $a_{k+1} = 3k + 1$.
4. [4] This is $3k + 1$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: D

Final answer

$a_n = 3n - 2$

Worked Solutions **Recurrence and Induction**

Q155 MCQ

Select the correct option: for $a_1 = 2$ and $a_{n+1} = a_n + 3$, the general term is:

Solution

1. [1] Base case: $a_1 = 2 = 2$.
2. [2] Assume $a_k = 3k - 1$ for some positive integer k.
3. [3] Substitute into $a_{k+1} = a_k + 3$: $a_{k+1} = 3k + 2$.
4. [4] This is $3k + 2$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: C

Final answer

$a_n = 3n - 1$

Worked Solutions **Recurrence and Induction**

Q156 MCQ

Select the correct option: for $a_1 = 3$ and $a_{n+1} = a_n + 3$, the general term is:

Solution

1. [1] Base case: $a_1 = 3 = 3$.
2. [2] Assume $a_k = 3k$ for some positive integer k.
3. [3] Substitute into $a_{k+1} = a_k + 3$: $a_{k+1} = 3(k+1)$.
4. [4] This is $3(k+1)$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: B

Final answer

$a_n = 3n$

Worked Solutions **Recurrence and Induction**

Q157 **MCQ**

Select the correct option: for $a_1 = 4$ and $a_{n+1} = a_n + 3$, the general term is:

Solution

1. [1] Base case: $a_1 = 4 = 4$.
2. [2] Assume $a_k = 3k + 1$ for some positive integer k.
3. [3] Substitute into $a_{k+1} = a_k + 3$: $a_{k+1} = 3k + 4$.
4. [4] This is $3k + 4$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: A

Final answer

$a_n = 3n + 1$

Worked Solutions **Recurrence and Induction**

Q158 Written

For a sequence with $a_1 = 1$ and $a_{n+1} = a_n^2 - n^2 a_n + (n+1)^2$, prove by mathematical induction that $a_n = n^2$.

Solution

1. [1] Base case: $a_1 = 1 = 1$.
2. [2] Assume $a_k = k^2$ for some positive integer k.
3. [3] Substitute into $a_{k+1} = {a_k}^2 - k^2 a_k + (k+1)^2$: $a_{k+1} = (k+1)^2$.
4. [4] This is $(k+1)^2$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Final answer

$a_n = n^2$

Worked Solutions **Recurrence and Induction**

Q159 **Written**

For a sequence with $a_1 = -1$ and $a_{n+1} = a_n^2 + 2na_n - 2$, prove by mathematical induction that $a_n = 1 - 2n$.

Solution

1. [1] Base case: $a_1 = -1 = -1$.
2. [2] Assume $a_k = 1 - 2k$ for some positive integer k.
3. [3] Substitute into $a_{k+1} = {a_k}^2 + 2ka_k - 2$: $a_{k+1} = -2k - 1$.
4. [4] This is $-2k - 1$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Final answer

$a_n = 1 - 2n$

Worked Solutions **Recurrence and Induction**

Q160 MCQ

Select the correct option: for $a_1 = 1$ and $a_{n+1} = 2a_n + 0$, the general term is:

Solution

1. [1] Base case: $a_1 = 1 = 1$.
2. [2] Assume $a_k = 2^{k-1}$ for some positive integer k.
3. [3] Substitute into $a_{k+1} = 2a_k + 0$: $a_{k+1} = 2^k$.
4. [4] This is 2^k, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: C

Final answer

$a_n = 2^{n-1}$

Worked Solutions **Recurrence and Induction**

Q161 Written

For a sequence with $a_1 = 1$ and $a_{n+1} = \frac{a_n - 4}{a_n - 3}$, prove by mathematical induction that $a_n = \frac{2n-1}{n}$.

Solution

1. [1] Base case: $a_1 = 1 = 1$.
2. [2] Assume $a_k = \frac{2k-1}{k}$ for some positive integer k.
3. [3] Substitute into $a_{k+1} = \frac{a_k - 4}{a_k - 3}$: $a_{k+1} = \frac{2k+1}{k+1}$.
4. [4] This is $\frac{2k+1}{k+1}$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Final answer

$$a_n = \frac{2n-1}{n}$$

Worked Solutions

Recurrence and Induction

Q162 Challenge

Three numbers A, B, C in that order form a geometric sequence with common ratio r. If $A, 2B, C$ in that order form an arithmetic sequence, determine the possible values of r.

Solution

1. Let the geometric sequence be A, Ar, Ar^2.
2. Since $A, 2B, C$ is arithmetic, its middle term is the average of the outer terms: $4B = A + C$.
3. Substitute $B = Ar, C = Ar^2$ and cancel the non-zero scale factor A: $r^2 - 4r + 1 = 0$.
4. $r = \frac{4 \pm \sqrt{16-4}}{2} = 2 \pm \sqrt{3}$.

Final answer

$r = 2 \pm \sqrt{3}$

Worked Solutions **Recurrence and Induction**

Quiz MCQ 1 Concept Quiz · MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 1a_n + (4)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 2(2n - 1)$.

Correct option: D

Final answer

$a_n = 2(2n - 1)$

Worked Solutions **Recurrence and Induction**

Quiz MCQ 2 **Concept Quiz · MCQ**

Select the correct option: If $a_1 = 2$ and $a_{n+1} = -2a_n + (3)$, the general term is:

Solution

1. $c = 1$ from $c = -2c + (3)$.
2. $a_{n+1} - c = -2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{(-2)^n - 2}{2}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: A

Final answer

$$a_n = -\frac{(-2)^n - 2}{2}$$

Quiz MCQ 3 Concept Quiz · MCQ

Select the correct option: which inequality is proved for $n \geq 4$?:

Solution

1. $n = 4: \quad 2^4 = 16 > 13 = 13.$
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $2^k + 1 > 2(3k + 1)$ after multiplying by the positive base.
4. $2(3k + 1) - (3(k + 1) + 1) = 3k - 2$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Correct option: B

Final answer

$2^n > 3n + 1$

Worked Solutions **Recurrence and Induction**

Quiz MCQ 4 **Concept Quiz · MCQ**

Select the correct option: for $a_1 = 4$ and $a_{n+1} = a_n + 2$, the general term is:

Solution

1. [1] Base case: $a_1 = 4 = 4$.
2. [2] Assume $a_k = 2\,(k+1)$ for some positive integer k.
3. [3] Substitute into $a_{k+1} = a_k + 2$: $a_{k+1} = 2\,(k+2)$.
4. [4] This is $2\,(k+2)$, the proposed formula with $n = k+1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: C

Final answer

$a_n = 2\,(n+1)$

Worked Solutions **Recurrence and Induction**

Quiz WRITTEN 5 **Concept Quiz · Written**

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 1a_n + (5)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 5n - 3$.

Final answer

$a_n = 5n - 3$

Worked Solutions **Recurrence and Induction**

Quiz WRITTEN 6 **Concept Quiz · Written**

Find the general term of $\{a_n\}$ given $a_1 = 1$ and $a_{n+1} = 5a_n + (8)$.

Solution

1. $c = -2$ from $c = 5c + (8)$.
2. $a_{n+1} - c = 5(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{3\,5^n - 10}{5}$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{3\,5^n - 10}{5}$$

Worked Solutions **Recurrence and Induction**

Quiz WRITTEN 7 **Concept Quiz · Written**

Prove by mathematical induction that $2^n > 4n + 1$ for $n \geq 4$.

Solution

1. $n = 4:\quad 2^4 = 16 > 17 = 17.$
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $2^k + 1 > 2(4k + 1)$ after multiplying by the positive base.
4. $2(4k + 1) - (4(k + 1) + 1) = 4k - 3$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Final answer

$2^n > 4n + 1$

Worked Solutions **Recurrence and Induction**

Quiz WRITTEN 8 **Concept Quiz · Written**

For a sequence with $a_1 = 5$ and $a_{n+1} = a_n + 1(2n+1) + 2$, prove by mathematical induction that $a_n = n^2 + 2n + 2$.

Solution

1. [1] Base case: $a_1 = 5 = 5$.
2. [2] Assume $a_k = k^2 + 2k + 2$ for some positive integer k.
3. [3] Substitute into $a_{k+1} = a_k + 1(2k+1) + 2$: $a_{k+1} = k^2 + 4k + 5$.
4. [4] This is $k^2 + 4k + 5$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Final answer

$a_n = n^2 + 2n + 2$

Answer key

Use the question number shown on each page.

Q001 $a_2 = 8,\ a_3 = 26,\ a_4 = 80$

Q002 $a_2 = -9,\ a_3 = -15,\ a_4 = -27$

Q003 $a_2 = -7,\ a_3 = -7,\ a_4 = -4$

Q004 $a_2 = -3,\ a_3 = -5,\ a_4 = -8$

Q005 $a_2 = 4,\ a_3 = 8,\ a_4 = 14$

Q006 $a_2 = 1,\ a_3 = 10,\ a_4 = 37$

Q007 B

Q008 D

Q009 A

Q010 B

Q011 D

Q012 A

Q013 C

Q014 D

Q015 C

Q016 D

Q017 B

Q018 C

Q019 $a_n = 5n - 6$

Q020 $a_n = 3n - 2$

Q021 $a_n = -2(n-3)$

Q022 $a_n = 6n - 5$

Q023 $a_n = 8 - 5n$

Q024 $a_n = 3n - 5$

Q025 $a_n = -4(n-2)$

Q026 D

Q027 A

Q028 A

Q029 A

Q030 $a_n = 2\,3^{n-1}$

Q031 $a_n = 3\,5^{n-1}$

Q032 $a_n = 4\,3^{n-1}$

Q033 $a_n = 2^{2n-2}$

Q034 B

Q035 B

Q036 D

Q037 A

Q038 $a_n = \dfrac{3(-2)^n}{2}$

Q039 $a_n = 2(-5)^{n-1}$

Q040 $a_n = \dfrac{(-4)^n}{2}$

Q041 B

Q042 C

Q043 B

Q044 C

Q045 $a_2 = 5,\ a_3 = 13,\ a_4 = 36$

Q046 $a_2 = 3,\ a_3 = -1,\ a_4 = 7$

Q047 $a_2 = 8,\ a_3 = 22,\ a_4 = 63$

Q048 $a_n = -\dfrac{4\,2^n - 3\,3^n - 1}{2}$

Q049 $a_n = (n-2)^2$

Q050 $a_n = \dfrac{3n^2 - 3n + 2}{2}$

Q051 $a_n = n^2 - n + 2$

Q052 $a_n = \dfrac{2n^3 - 3n^2 + n + 4}{2}$

Q053 $a_n = \dfrac{3n^2 - n + 2}{2}$

Q054 $a_n = \dfrac{4n^3 - 3n^2 - n + 12}{6}$

Q055 C

Q056 A

Q057 C

Q058 D

Q059 C

Q060 A

Q061 D

Q062 A

Q063 B

Q064 A

Q065 $a_n = \dfrac{2n^3 - 3n^2 + n + 12}{6}$

Q066 A

Q067 $a_n = 2^n - 1$

Q068 $a_n = -\dfrac{3^n - 7}{2}$

Q069 $a_n = -\dfrac{4^n - 16}{3}$

Q070 $a_n = -\dfrac{n^2 - n - 200}{50}$

Q071 $a_n = 3^n - 2$

Q072 $a_n = \dfrac{3\,5^n - 10}{5}$

Q073 $a_n = \dfrac{2^n + 2}{2}$

Q074 $a_n = -\dfrac{4^n + 2}{2}$

Q075 $a_n = \dfrac{2\,3^n + 3}{3}$

Q076 $a_n = (-3)^n + 1$

Q077 $a_n = 3\,3^{-n}(3^n - 2)$

Q078 $a_n = \dfrac{3\,3^{-n}(7\,2^n - 2\,3^n)}{4}$

Q079 $a_n = -\dfrac{3(-2)^n - 2}{2}$

Q080 B

Q081 C

Q082 D

Q083 B

Q084 C

Q085 D

Q086 B

Q087 C

Q088 D

Q089 D

Q090 $c = 3, \quad a_n = -\dfrac{2^{2n} - 12}{4}, \quad a_5 = -253$

Q091 $\dfrac{n(n+1)}{2}$

Q092 $\dfrac{n(2n-1)(2n+1)}{3}$

Q093 $\dfrac{n(n+1)(n+2)}{3}$

Q094 A

Q095 B

Q096 C

Q097 B

Q098 C

Q099 A

Q100 C

Q101 A

Q102 B

Q103 n^2

Q104 $\frac{n(n+1)(n+2)}{3}$

Q105 $\frac{n(n+1)(2n+7)}{6}$

Q106 $4^n - 1$

Q107 $\frac{10\,10^n - 1}{9}$

Q108 $\frac{n\left(n^2+3n+5\right)}{3}$

Q109 $2^n > 3n$

Q110 $3^n > 5n + 1$

Q111 $3^n > 2n$

Q112 $2^n > 1n + 2$

Q113 $3^n > 5n + 1$

Q114 $3^n > 5n + 2$

Q115 $4^n > 2n + 3$

Q116 D

Q117 A

Q118 B

Q119 D

Q120 A

Q121 B

Q122 D

Q123 A

Q124 $2^n > 3n + 1$

Q125 B

Q126 $(1+h)^n > 1 + nh^2$

Q127 $9 \mid [n^3 + (n+1)^3 + (n+2)^3]$

Q128 $3 \mid (n^3 + 2n)$

Q129 $2 \mid (n^2 + n)$

Q130 $2 \mid (n^2 + 5n + 8)$

Q131 $2 \mid (n^2 + 7n + 12)$

Q132 $2 \mid (n^2 + 11n + 32)$

Q133 A

Q134 D

Q135 C

Q136 C

Q137 B

Q138 A

Q139 D

Q140 D

Q141 D

Q142 $6 \mid (2n^3 - 3n^2 + n)$

Q143 $3 \mid (4n^3 - n)$

Q144 $n^2 \mid [(n+1)^n - 1]$

Q145 $a_2 = 4, \quad a_3 = 5, \quad a_4 = 6; \quad a_n = n + 2$

Q146 $a_2 = -3, \quad a_3 = -5, \quad a_4 = -7; \quad a_n = 1 - 2n$

Q147 $a_2 = 4, \quad a_3 = 9, \quad a_4 = 16; \quad a_n = n^2$

Q148 $a_2 = \frac{3}{2}, \quad a_3 = \frac{5}{3}, \quad a_4 = \frac{7}{4}; \quad a_n = \frac{2n-1}{n}$

Q149 $a_n = n + 2$

Q150 D

Q151 C

Q152 B

Q153 A

Q154 D

Q155 C

Q156 B

Q157 A

Q158 $a_n = n^2$

Q159 $a_n = 1 - 2n$

Q160 C

Q161 $a_n = \frac{2n-1}{n}$

Q162 $r = 2 \pm \sqrt{3}$

Quiz MCQ 1 D

Quiz MCQ 2 A

Quiz MCQ 3 B

Quiz MCQ 4 C

Quiz WRITTEN 5 $a_n = 5n - 3$

Quiz WRITTEN 6 $a_n = \frac{3\,5^n - 10}{5}$

Quiz WRITTEN 7 $2^n > 4n + 1$

Quiz WRITTEN 8 $a_n = n^2 + 2n + 2$